



Started on	Wednesday, 30 September 2026, 10:30 PM
State	Finished
Completed on	Wednesday, 30 September 2026, 10:30 PM
Time taken	51 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      int arr[n];
8
9      for (int i = 0; i < n; i++) {
10         scanf("%d", &arr[i]);
11     }
12
13     for (int i = 0; i < n; i++) {
14         for (int j = i + 1; j < n; j++) {
15             if (arr[i] == arr[j]) {
16                 printf("%d", arr[i]);
17                 return 0;
18             }
19         }
20     }
21
22     return 0;
23 }
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



Started on	Wednesday, 30 September 2026, 10:31 PM
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Completed on	Wednesday, 30 September 2026, 10:31 PM
Time taken	30 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Find Duplicate in Array.

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14         for (int j = i + 1; j < n; j++) {
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18             }
19         }
20     }
21
22     return 0;
23 }
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

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Correct

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[Back to Course](#)



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Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int T;
5      scanf("%d", &T);
6
7      while (T--) {
8          int n1, n2;
9
10         scanf("%d", &n1);
11
12         int a[n1];
13
14         for (int i = 0; i < n1; i++) {
15             scanf("%d", &a[i]);
16         }
17
18         scanf("%d", &n2);
19
20         int b[n2];
21
22         for (int i = 0; i < n2; i++) {

```

```
23     scanf("%d", &b[i]);
24 }
25
26 int i = 0, j = 0;
27
28 while (i < n1 && j < n2) {
29     if (a[i] == b[j]) {
30         printf("%d ", a[i]);
31         i++;
32         j++;
33     }
34     else if (a[i] < b[j]) {
35         i++;
36     }
37     else {
38         j++;
39     }
40 }
41
42 printf("\n");
43 }
44
45 return 0;
46 }
```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

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[Back to Course](#)



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Question 1 | Correct | Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```

1 #include <stdio.h>
2
3 int main() {
4     int T;
5     scanf("%d", &T);
6
7     while (T--) {
8         int n1, n2;
9
10        scanf("%d", &n1);
11        int a[n1];
12
13        for (int i = 0; i < n1; i++)
14            scanf("%d", &a[i]);
15
16        scanf("%d", &n2);
17        int b[n2];
18
19        for (int i = 0; i < n2; i++)
20            scanf("%d", &b[i]);
21
22        int i = 0, j = 0;
```

```
23
24 while (i < n1 && j < n2) {
25     if (a[i] == b[j]) {
26         printf("%d ", a[i]);
27         i++;
28         j++;
29     }
30     else if (a[i] < b[j]) {
31         i++;
32     }
33     else {
34         j++;
35     }
36 }
37
38 printf("\n");
39 }
40
41 return 0;
42 }
```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

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Back to Course



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Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n, k;
5      scanf("%d", &n);
6
7      int a[n];
8
9      for (int i = 0; i < n; i++) {
10         scanf("%d", &a[i]);
11     }
12
13     scanf("%d", &k);
14
15     int i = 0, j = 1;
16
17     while (i < n && j < n) {
18         if (i == j) {
19             j++;
20         }
21         else if (a[j] - a[i] == k) {
22             printf("1");
23             return 0;
24         }
25         else if (a[j] - a[i] < k) {
26             j++;
27         }
28         else {
29             i++;
30         }
31     }
32
33     printf("0");
34
35     return 0;
36 }
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

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Back to Course



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Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3	1
1 3 5	
4	

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n, k;
5      scanf("%d", &n);
6
7      int a[n];
8
9      for (int i = 0; i < n; i++) {
10         scanf("%d", &a[i]);
11     }
12
13     scanf("%d", &k);
14
15     int i = 0, j = 1;
16
17     while (i < n && j < n) {
18         if (i == j) {
19             j++;
20         } else if (a[j] - a[i] == k) {
21             printf("1");
22             return 0;
23         } else if (a[j] - a[i] < k) {
24             j++;
25         } else {
26             i++;
27         }
28     }
29
30     printf("0");
31
32     return 0;
33 }
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

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[Back to Course](#)